

Esquema de subredes utilizadas

Sucursal 1

Subred	Dirección de Red	Máscara de Subred	Gateway	Rango de IPs Utilizables	Dirección de Broadcast	Hosts Disponibles
<i>Directivos</i>	10.20.136.0	/23 (255.255.254.0)	10.20.136.1	10.20.136.1 - 10.20.137.254	10.20.137.255	510
<i>Enfermería</i>	10.20.138.0	/23 (255.255.254.0)	10.20.138.1	10.20.138.1 - 10.20.139.254	10.20.139.255	510
<i>RRHH</i>	10.20.134.0	/23 (255.255.254.0)	10.20.134.1	10.20.134.1 - 10.20.135.254	10.20.135.255	510
<i>Médicos</i>	10.20.132.0	/23 (255.255.254.0)	10.20.132.1	10.20.132.1 - 10.20.133.254	10.20.133.255	510
<i>WiFi</i>	10.20.130.0	/23 (255.255.254.0)	10.20.130.1	10.20.130.1 - 10.20.131.254	10.20.131.255	510
<i>Servidores</i>	10.20.128.0	/23 (255.255.254.0)	10.20.128.1	10.20.128.1 - 10.20.129.254	10.20.129.255	510

Sucursal 2

Subred	Dirección de Red	Máscara de Subred	Gateway	Rango de IPs Utilizables	Dirección de Broadcast	Hosts Disponibles
<i>Directivos</i>	192.168.110.128	/27 (255.255.255.224)	192.168.110.129	192.168.110.129 - 192.168.110.158	192.168.110.159	30
<i>Enfermería</i>	192.168.110.160	/27 (255.255.255.224)	192.168.110.161	192.168.110.161 - 192.168.110.190	192.168.110.191	30
<i>RRHH</i>	192.168.110.96	/27 (255.255.255.224)	192.168.110.97	192.168.110.97 - 192.168.110.126	192.168.110.127	30
<i>Médicos</i>	192.168.110.64	/27	192.168.110.65	192.168.110.6	192.168.110	30

		(255.255.255.224)		5 - 192.168.110.9 4	.95	
<i>WiFi</i>	192.168.110.32	/27 (255.255.255.224)	192.168.110.33	192.168.110.3 3 - 192.168.110.6 2	192.168.110 .63	30
<i>Servidores</i>	192.168.110.0	/27 (255.255.255.224)	192.168.110.1	192.168.110.1 - 192.168.110.3 0	192.168.110 .31	30

Sucursal 3

Subred	Dirección de Red	Máscara de Subred	Gateway	Rango de IPs Utilizables	Dirección de Broadcast	Hosts Disponibles
<i>Directivos</i>	172.30.0.0	/25 (255.255.255.128)	172.30.0.1	172.30.0.1 - 172.30.0.126	172.30.0.12 7	126
<i>Enfermería</i>	172.30.128.0	/25 (255.255.255.128)	172.30.128. 1	172.30.128.1 - 172.30.128.254	172.30.128. 255	126
<i>RRHH</i>	172.30.96.0	/25 (255.255.255.128)	172.30.96.1	172.30.96.1 - 172.30.96.126	172.30.96.1 27	126
<i>Médicos</i>	172.30.64.0	/25 (255.255.255.128)	172.30.64.1	172.30.64.1 - 172.30.64.126	172.30.64.1 27	126
<i>WiFi</i>	172.30.160.0	/25 (255.255.255.128)	172.30.160. 1	172.30.160.1 - 172.30.160.126	172.30.160. 127	126
<i>Servidores</i>	172.30.32.0	/25 (255.255.255.128)	172.30.32.1	172.30.32.1 - 172.30.32.126	172.30.32.1 27	126

IPs asignadas a servidores

Servicio	Dirección IP
<i>DHCP</i>	192.168.110.2
<i>DNS</i>	172.30.32.6
<i>FTP</i>	10.20.128.2

<i>SMTP</i>	10.20.128.4
<i>Web Institucional</i>	192.168.110.3
<i>Web Directivos</i>	192.168.110.5
<i>Web Médicos</i>	192.168.110.6
<i>Web RRHH</i>	192.168.110.4

Usuarios y contraseñas del servidor SMTP

PC/Laptop	Usuario	Rol	Email	Usuario FTP	Contraseña
RRHH	rrhh1	RW	rrhh1@clinica.org	rrhh1	123
RRHH2	rrhh2	RW	rrhh2@clinica.org	rrhh2	123
Suc2_RRHH	rrhh3	RW	rrhh3@clinica.org	rrhh3	123
Suc2_RRHH_2	rrhh4	RW	rrhh4@clinica.org	rrhh4	123
Sucursal3_RRHH	rrhh5	RW	rrhh5@clinica.org	rrhh5	123
Sucursal3_RRHH_2	rrhh6	RW	rrhh6@clinica.org	rrhh6	123
Médico	medico1	RW	medico1@clinica.org	medico1	123
Médico2	medico2	RW	medico2@clinica.org	medico2	123
Suc2_Medico	medico3	RW	medico3@clinica.org	medico3	123
Suc2_Medico2	medico4	RW	medico4@clinica.org	medico4	123
Sucursal3_Medico	medico5	RW	medico5@clinica.org	medico5	123
Sucursal3_Medico_2	medico6	RW	medico6@clinica.org	medico6	123
Directivo	dir1	RW	dir1@clinica.org	dir1	123
Directivo2	dir2	RW	dir2@clinica.org	dir2	123
Suc2_Directivo	dir3	RW	dir3@clinica.org	dir3	123
Suc2_Directivo2	dir4	RW	dir4@clinica.org	dir4	123
Sucursal3_Directivo	dir5	RW	dir5@clinica.org	dir5	123
Sucursal3_Directivo_2	dir6	RW	dir6@clinica.org	dir6	123

Enfermería	enfer1	L	enfer1@clinica.org	enfer1	123
Enfermería2	enfer2	L	enfer2@clinica.org	enfer2	123
Suc2_Enfermería	enfer3	L	enfer3@clinica.org	enfer3	123
Suc2_Enfermería2	enfer4	L	enfer4@clinica.org	enfer4	123
Sucursal3_Enfermeria	enfer5	L	enfer5@clinica.org	enfer5	123
Sucursal3_Enfermeria_2	enfer 6	L	enfer6@clinica.org	enfer6	123
Suc1_Pb_Medico_Laptop	wifi1	RDL	wifi1@clinica.org	wifi1	123
Suc1_Pb_RRHH_Laptop	wifi2	RDL	wifi2@clinica.org	wifi2	123
Suc1_1ro_Medico_Laptop	wifi3	RDL	wifi3@clinica.org	wifi3	123
Suc1_1ro_RRHH_Laptop	wifi4	RDL	wifi4@clinica.org	wifi4	123
Suc2_Laptop_RRHH2	wifi5	RDL	wifi5@clinica.org	wifi5	123
Suc2_Laptop_Medico2	wifi6	RDL	wifi6@clinica.org	wifi6	123
Suc2_Laptop_RRHH	wifi7	RDL	wifi7@clinica.org	wifi7	123
Suc2_Laptop_Medico	wifi8	RDL	wifi8@clinica.org	wifi8	123
Sucursal3_Pb_Laptop_RRHH_2	wifi9	RDL	wifi9@clinica.org	wifi9	123
Sucursal3_Pb_Laptop_Medico_2	wifi10	RDL	wifi10@clinica.org	wifi10	123
Sucursal3_1ro_Laptop_RRHH	wifi11	RDL	wifi11@clinica.org	wifi11	123
Sucursal3_1ro_Laptop_Medico	wifi12	RDL	wifi12@clinica.org	wifi12	123

Listado utilizado de comandos para configuración de Routers

```

ip dhcp excluded-address 10.20.136.1
ip dhcp excluded-address 10.20.138.1
ip dhcp excluded-address 10.20.130.1
ip dhcp excluded-address 10.20.134.1
ip dhcp excluded-address 10.20.132.1
ip dhcp excluded-address 10.20.128.1
ip dhcp excluded-address 10.20.128.3

```

```
ip dhcp pool directivos
network 10.20.136.0 255.255.254.0
default-router 10.20.136.1
```

```
dns-server 172.30.32.6
ip dhcp pool enfermeros
network 10.20.138.0 255.255.254.0
default-router 10.20.138.1
dns-server 172.30.32.6
```

```
ip dhcp pool wifi
network 10.20.130.0 255.255.254.0
default-router 10.20.130.1
dns-server 172.30.32.6
```

```
ip dhcp pool rrhh
network 10.20.134.0 255.255.254.0
default-router 10.20.134.1
dns-server 172.30.32.6
```

```
ip dhcp pool medicos
network 10.20.132.0 255.255.254.0
default-router 10.20.132.1
dns-server 172.30.32.6
```

```
ip dhcp pool servidores
network 10.20.128.0 255.255.254.0
default-router 10.20.128.1
dns-server 172.30.32.6
```

```
interface GigabitEthernet0/0/0.10
encapsulation dot1Q 10
ip address 10.20.136.1 255.255.254.0
```

```
interface GigabitEthernet0/0/0.20
encapsulation dot1Q 20
ip address 10.20.138.1 255.255.254.0
```

```
interface GigabitEthernet0/0/0.30
encapsulation dot1Q 30
ip address 10.20.130.1 255.255.254.0
```

```
interface GigabitEthernet0/0/0.40
encapsulation dot1Q 40
```

```
ip address 10.20.134.1 255.255.254.0
```

```
interface GigabitEthernet0/0/0.50  
encapsulation dot1Q 50  
ip address 10.20.132.1 255.255.254.0
```

```
interface GigabitEthernet0/0/0.60  
encapsulation dot1Q 60  
ip address 10.20.128.1 255.255.254.0
```

```
interface Serial0/1/0  
ip address 8.157.126.250 255.255.255.252
```

```
interface Serial0/1/1  
ip address 8.157.126.241 255.255.255.252
```

```
router rip  
network 8.0.0.0  
network 10.0.0.0
```

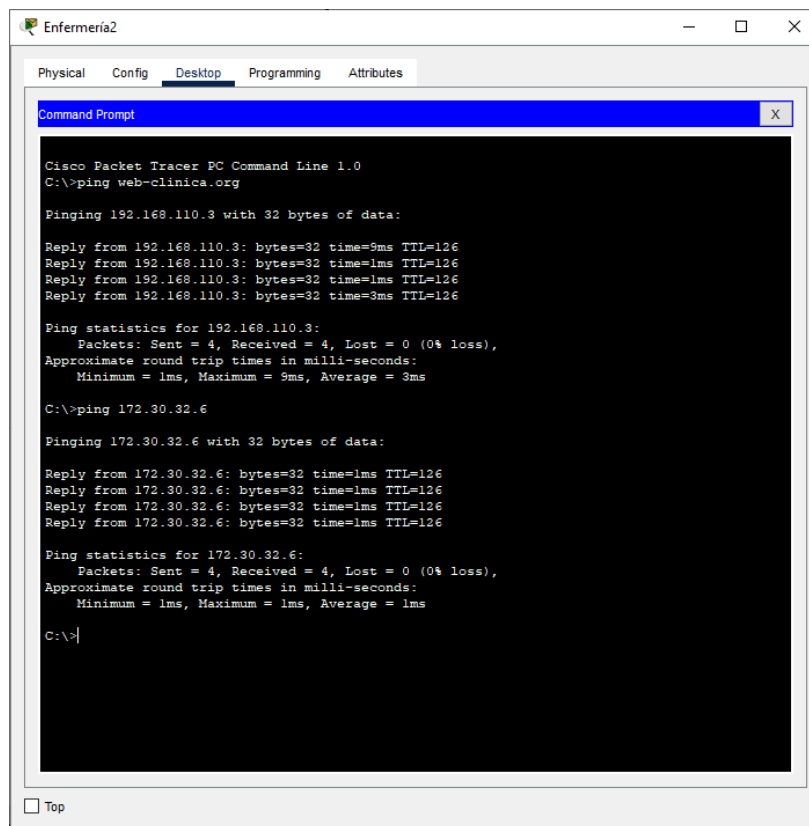
Tabla de Enrutamiento

Direcciones IP	Máscara	(Estático/RIP)	Destino	Origen
8.157.126.250	255.255.255.252	RIP	Router 3	Router 1
8.157.126.241	255.255.255.252	RIP	Router 2	Router 1
8.157.126.242	255.255.255.252	RIP	Router 1	Router 2
8.157.126.245	255.255.255.252	RIP	Router 3	Router 2
8.157.126.249	255.255.255.252	RIP	Router 1	Route 3
8.157.126.246	255.255.255.252	RIP	Router 2	Route 3

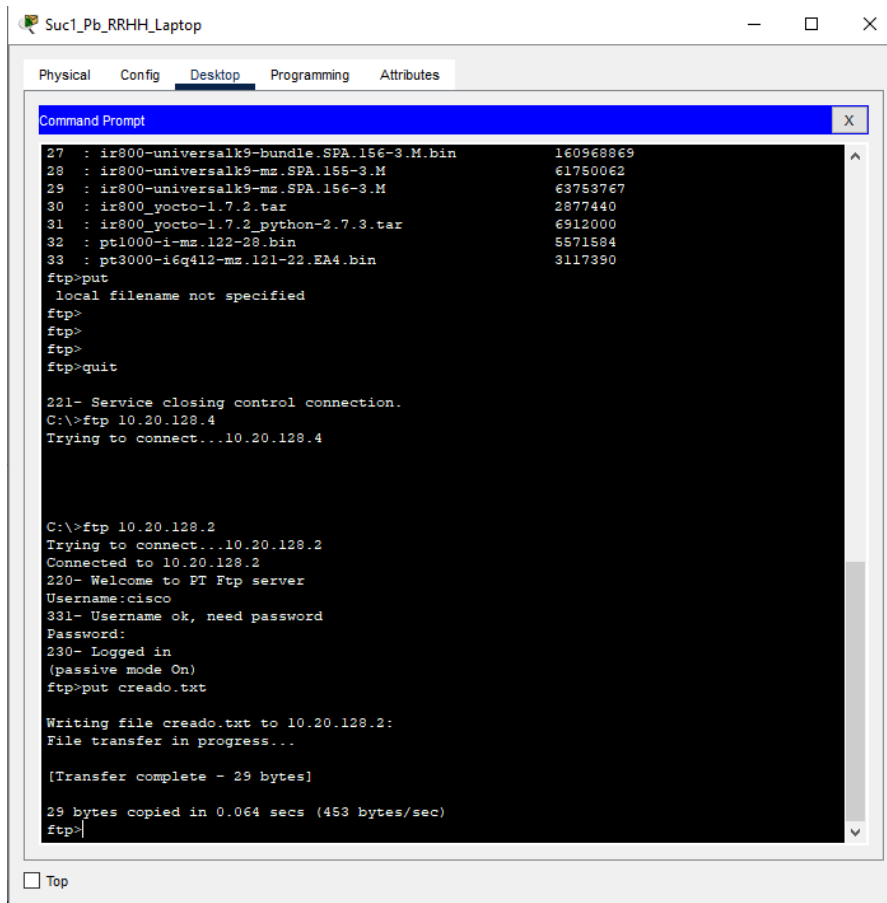
Usando la herramienta de PING gráfica [p]

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Directivo2	Suc2_RRHH_2	ICMP		0.000	N	0	(edit)	

PING por consola yendo a la PC → Desktop → Command Prompt →
ping [IP o DNS]



Salto de paquetes usando el comando Command Prompt



Prueba Navegación Web

